

2018

Time : 3 hours

Full Marks : 80

*Candidates are required to give their answers in
their own words as far as practicable.*

The questions are of equal value.

*Answer any **five** questions.*

1. Discuss OPAMA as an adder and subtractor.
2. Explain working of TTL – NOR gate and discuss its input and output margins.
3. What is meant by race – around condition ? How it is removed in master slave flip-flops ?
4. Design a four bit synchronous counter and discuss its working.

5. State and explain Demorgan's theorem. Using theorem prove that $AB + \bar{A}\bar{B} = \overline{(\bar{A}\bar{B} + \bar{A}\bar{B})}$
6. Give circuit diagram of analogue to digital converter and discuss for parallel 3-bit.
7. Design Logarithmic and anti-logarithmic amplifiers and discuss.
8. Draw a diagram showing how RAM, ROMs and ports are added to a CPU to make simple microcomputer.
9. Describe how many address lines are required by a microprocessors to address 4K memory.
10. Write short notes on any **two** of the following :
 - (a) Demultiplexer
 - (b) Ladder network
 - (c) Square wave generator
 - (d) NAND gate as universal gate

